Answer key

TMUA October 2026 - reconstructed 40-question paper

These letters refer to the numbering and options in the accompanying compiled paper.

Paper 1		Paper 2	
Question	Answer	Question	Answer
1	E	1	C
2	F	2	C
3	B	3	B
4	A	4	D
5	A	5	F
6	D	6	B
7	B	7	A
8	D	8	B
9	D	9	F
10	D	10	G
11	C	11	E
12	B	12	C
13	D	13	B
14	D	14	D
15	B	15	B
16	E	16	A
17	C	17	C
18	D	18	C
19	C	19	C
20	D	20	C

Scoring: one mark per correct answer; 20 marks per paper, 40 in total.

Paper 1: answer checks

1. The six roots are $-\pi/6, 0, \pi/3, 5\pi/6, \pi, 4\pi/3$, with sum $10\pi/3$.

2. The fifth and sixth positive roots of $\sin u = 1/2$ are $25\pi/6$ and $29\pi/6$. Hence $25/6 \leq k < 29/6$.

3. The chord PQ is $y = 15x + 20$. Maximise $21x - x^3 - 20$ on $(1, 4)$ to obtain $x = \sqrt{7}$.

4. The repeated root is -3 or $-1/3$, giving $c = -9$ or $13/27$. Their sum is $-230/27$.

5. The angle-bisector theorem and sine rule give $RT/TS = QR/QS = \sin S/\sin R$.

6. For $k \neq 0$, intersection requires $3(x - k/2)^2 + (k^2 - 9)/4 = 0$. Include $k = 0$, when the curves coincide.

7. The remainder conditions give $2p - q = 4$ and $p + q = 11$. Thus $p = 5$, $q = 6$, and $2p + 13q = 88$.

8. The centre is distance 2 from the line. The minor central angle is $2\pi/3$, giving arc length $8\pi/3$.

9. Grouping odd and even powers gives $11111\sqrt{10} + 11110$.

10. With $t = x - 2$, $f(x) = t^4/4 - t^2/2 + C$. Only the level C has exactly three roots; hence $a = b = C$.

11. Let $A = \sqrt{x} + \sqrt{x+1}$. Subtract the reciprocal relation to obtain $2\sqrt{y} = A^2 - A^{-2} = 4\sqrt{x(x+1)}$.

12. The graph is always positive, has period 2π, passes through $(0, 1)$ increasing, and has range $[1/3, 3]$.

13. For $0 < x < 1$, $2^{|\log_2 x|} = 1/x$; for $x \geq 1$, it equals x.

14. Write $x^2 + x - 12 = k\pi$. The cases $k = -3, -2, -1, 0$ give respectively $2, 2, 1, 1$ roots in $(-\pi, \pi)$.

15. On this interval, the second derivative is negative for I and positive for II and III. Only I is underestimated.

16. The expression is $1 - 2x$ for $x \leq 1/2$, $2x - 1$ for $1/2 < x < 1$, and 1 for $x \geq 1$.

17. The fractional-part integral is $5/2$, so the result is $2(5/2) - 25/2 = -15/2$.

18. The roots are $3 \pm \sqrt{3 + \sqrt{3}}$ and $3 \pm \sqrt{3 - \sqrt{3}}$: four distinct values.

19. Put $h(x) = (x - 1)^2$. Inverting $h^{\circ 4}(x) = 1$ gives $h(x) \in \{0, 2, 1 + \sqrt{1 + \sqrt{2}}\}$, hence five roots.

20. If $b < 0$, then $a^2 > b^2$; if $b \geq 0$, then $c^2 > b^2$. Both inequalities can also hold.

Paper 2: answer checks

1. Independence requires $n = 7$. Each of the six values of m then works.

2. Writing $t = \log_x y > 0$, the second inequality becomes $3t/(1+t) > 2$, equivalent to $t > 2$.

3. The square areas form $8 + 2 + 1/2 + \cdots$, with sum $32/3$.

4. $6 \mid 8 \cdot 9$, but $6 \nmid 8$ and $6 \nmid 9$.

5. If $I(t) = \int_{-t}^{t} f(x)\,dx \le 0$ for all t, then $I(-t) = -I(t)$ forces $I(t) = 0$ for all t.

6. There are two positive roots, so $b^2 - 4ac > 0$, $a, c > 0$ and $b < 0$. The sketch does not determine $f(1)$.

7. Q holds exactly when at least two of a, b, c are even, so $Q \Rightarrow P$. The converse fails for $(4, 1, 1)$.

8. I is $(8 \cdot 63!)^2$. In II, the prime 61 has odd exponent; III is $65(64!)^2$, also not a square.

9. Start with $\int_a^d (g - f)\,dx$ and reverse its sign on (b, c) by adding $2\int_b^c (f - g)\,dx$.

10. The first identity holds exactly when $p = \pi/2 + 2j\pi$, not at every odd multiple of $\pi/2$. Its converse and T are true.

11. $f(x) = 3x^2 - 4x + 1$, so $f'(x) = 6x - 4 > 0$ exactly when $x > 2/3$.

12. At $k = 0$ both lines are $y = x$. Otherwise $2^k \neq 1$, so the two slopes differ and there is one intersection.

13. $S = na + dn(n-1)/2$. If d is odd, the parity may depend on n modulo 4, which the value of n determines.

14. The maximum of $2x + y$ on the circle is $2p + q + 5$. Therefore $2p + q \le 5$, with equality attainable.

15. Choose 6 and two of $1, 5, 7$ in 3 ways, or one of $2, 4, 8, 10$, one of $3, 9$, and one of $1, 5, 7$ in 24 ways.

16. For $x \neq 0$, the sum is $x/|x|$; for $x = 0$, it is 0. Thus the set is $\{-1, 0, 1\}$.

17. The working is invalid at step II. The original equation gives $\log_2 x = 2$ or 3, so $x = 4$ or 8.

18. The first copy gives $b = -4$, and the second gives $c = -12$. Hence the original roots are 6 and -2.

19. Inclusion-exclusion gives $\mathrm{P}(2) = 0.4 + 0.7 - (1 - 0.2) = 0.3$.

20. IV gives $c = 2b - a$; with III, the triangle inequality $a + b > c$ is equivalent to I, namely $2a > b$.

Compilation notes

Question count and provenance

The supplied `TMUA_2026 (2).pdf` contains eight Paper 1 questions and nineteen Paper 2 questions, for a total of 27. All are retained in their original relative order. Reaching 20 questions per paper requires twelve additions to Paper 1 and one to Paper 2.

The added questions come from the supplied Qiuyun compilation. Its question numbers and physical PDF pages are listed below. Incomplete recollections were excluded.

Compiled paper	Qiuyun question	Source PDF page
Paper 1, Q9	2	3
Paper 1, Q10	3	4
Paper 1, Q11	4	5
Paper 1, Q12	5	6
Paper 1, Q13	6	7
Paper 1, Q14	7	8
Paper 1, Q15	8	9
Paper 1, Q16	9	10
Paper 1, Q17	11	12
Paper 1, Q18	12	13
Paper 1, Q19	13	14
Paper 1, Q20	22	23
Paper 2, Q20	27	28

Editorial changes

- Added questions without supplied options were adapted to multiple-choice format. Paper 1 Q19 retains the five options supplied with its source question.
- Paper 1 Q3: corrected "between P and R" to "between P and Q".
- Paper 2 Q14: corrected the repeated option label A to an A–E sequence. The answer 5 is therefore option D in this edition.
- Paper 2 Q15: specified three *distinct* integers. Allowing repetition would give 30, which is absent from the supplied options.
- Paper 2 Q16: corrected the reversed interval endpoints in options C and D to $(-1, 1)$ and $[-1, 1]$.
- Other edits clarify notation, punctuation and implicit domains. The coefficient $2/\sqrt{3}$ in Paper 1 Q1 and the slope 2^k in Paper 2 Q12 follow the rendered base PDF.